IS THE HOT HAND REAL? EVIDENCE FROM A PERMUTATION AND HIERARCHICAL-BASED ANALYSIS

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Cameron An

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TITLE:	Is the Hot Hand Real? Evidence from a Permutation and Hierarchical-Based Analysis
AUTHOR:	Cameron An
DATE SUBMITTED:	June 2026
COMMITTEE CHAIR:	Kevin Ross, Ph.D. Professor of Statistics
COMMITTEE MEMBER:	Bret Holladay, Ph.D. Assistant Professor of Statistics
COMMITTEE MEMBER:	Immanuel Williams, Ph.D. Lecturer of Statistics

ABSTRACT

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Cameron An

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Chapter 1

INTRODUCTION

There will probably be lots of citations here for many articles or even software like R (R Core Team, 2024).

The concept of the hot hand in basketball has been fiercely debated among audiences spanning many decades. Fans and players often describe the phenomenon as reaching an elevated state of play following a streak of recent success. One general notion of hot-hand shooting posits that players tend to make a higher percentage of shots relative to their baseline shooting ability following a recent streak of made shots. Proponents of the hot hand argue that the physical and psychological benefits of witnessing a streak of success place players “in rhythm”, a state of increased confidence and ability that may increase the chances of making a shot. Former Golden State Warriors player Purvis Short described the experience of being “hot” as follows: “You’re in a world all your own. It’s hard to describe. But the basket seems to be so wide. No matter what you do, you know the ball is going in” (Short, as quoted in Tversky & Gilovich, 1a 989). However, opponents argue that these experiences may reflect cognitive biases, suggesting that apparent streaks may arise purely from random variation. These perspectives highlight the intuitive appeal of the hot-hand phenomenon, raising the question of whether perceived streaks in performance reflect underlying changes in ability under unique circumstances or simply arise from chance.

Chapter 2

BACKGROUND

2.1 Previous Literature

Statistical research has questioned whether the hot hand truly exists, despite widespread belief from fans, players, and the general public. In a seminal study, Gilovich, Vallone, and Tversky (1985), examined streak-shooting tendencies to determine if observed streaks exceeded those expected under randomness. The sample data consisted of field goal attempts from 48 home games of 76ers players and their opponents during the 1980-81 season. The first section of their analysis used conditional proportions to examine players' shooting percentages conditional on the outcome of previous shots. For each of nine players on the 76ers, Gilovich, Vallone, and Tversky computed the overall shooting proportion as well as conditional shooting proportions based on whether the previous one, two, or three shots were successful. They also estimated the serial correlation between the outcomes of successive shots to determine if there existed any dependence structure within a player's shot sequence. A "success", or "hit", was defined as a made shot. The authors discovered that only one out of the nine sampled players possessed a higher success rate on shots following a make compared to their overall success rate. Furthermore, eight out of nine players exhibited a negative serial correlation value, none of which being significantly different from zero.

The findings in this paper contributed to the characterization of the hot hand as a myth, which researchers attribute to a cognitive fallacy whereby individuals misinterpret random streaks as performance trends (Gilovich et al., 1985). A potential explanation for such a phenomenon involves memory bias, the idea that observers are likely to overestimate the dependence of sequential shots due to long sequences of successes (or failures) being more memorable than alternating successes and failures (Gilovich et al., 1985). Other studies have since expanded upon the analysis in Gilovich et al., (1985), yielding similar results of a lack of hot-hand effect (Koehler & Conley, 2003; McNair et al., 2020; Lantis & Nesson, 2021).

However, recent research has revisited the validity of the methods used in Gilovich et al. (1985). In particular, Miller and Sanjurjo (2018) identified a flaw in the conditional proportion analysis used in the 1985 paper. The flaw stems from a systematic bias in the analysis of conditional shooting percentages within finite shot sequences, which ultimately resulted in an underestimation of the true hot hand effect. Miller & Sanjurjo (2018) formally demonstrated the downward bias in conditional proportion estimators by analyzing all possible permutations of short finite sequences of independent binary outcomes. In their example using three consecutive coin flips, they examine the proportion of heads following a previous head under the assumption of independence. A calculation of the conditional proportion estimator revealed that the expected proportion of heads conditioned on a previous head is equal to $\frac{5}{12}$, notably lower than the suspected 0.5, illustrating the systematic downward bias introduced by conditioning on finite sequences (Miller & Sanjurjo, 2018).

Miller and Sanjurjo applied the coin-flipping framework to demonstrate bias in the methodology of Gilovich et al. (1985), whereby the conditional proportions systematically underestimated the true hot hand effect of the data. Coin flips that resulted in heads were treated as made shots, while coin flips that resulted in tails were treated as missed shots. Applying the same logic to a basketball context, Miller and Sanjurjo reasoned that finite samples of shot data exhibited the same bias observed in the coin problem. More specifically, the expected proportion of made shots conditioned on a previous make was not equal to the player's overall shooting percentage, but instead was biased downwards due to the structure of the sampling process. To fix this issue, Miller and Sanjurjo utilized a permutation test, descried in detail in Section 4.1. After accounting for the downward bias in conditional proportions with the permutation test, Miller and Sanjurjo discovered a reversal of Gilovich's findings, indicating a discernible hot-hand effect for the given data.

2.2 Contextual Variation in Hot-Hand Evidence

While much research has been conducted on hot-hand trends, the settings of each analysis provide little overlap and, therefore, are applied in different contexts. In addition to Gilovich's analysis of 76ers players during the 1980-81 season, an analysis of conditional proportions under a controlled shooting experiment with the Cornell intercollegiate men's and women's basketball teams revealed that the outcomes of shot attempts were largely independent of the outcome of the previous attempt (Gilovich et al., 1985). The controlled experiment required players to shoot under highly regulated conditions, along with monetary incentivization determined by the accuracy of their shots. These results were ultimately subject to the same downward bias introduced by Miller and Sanjurjo.

Other analyses have yielded different results utilizing alternative methods and settings. An analysis of serial dependence on 110,513 free throws in the NBA revealed a discernible, albeit small, hot-hand effect in free throws (Mews & Ötting, 2023). This analysis accounted for unevenly spaced time points between free throws by using a state-space model that assesses a player's form using the Ornstein-Uhlenbeck process. Data from the NBA Three-Point Contest also provide evidence of a hot-hand effect using the permutation testing procedure presented by Miller and Sanjurjo (Ross, 2017). The aforementioned analyses attempt to isolate the hot-hand effect by considering shots under controlled conditions to prevent additional covariates from influencing the results. While this approach yields strong results, they may not generalize to real game situations.

Additional research has offered unique perspectives on hot-hand trends by incorporating context into a player's shots. An analysis of over 83,000 real game shots from the 2012-13 NBA season revealed that players shooting over their baseline shooting percentage in recent shots face different circumstances compared to when they are in their baseline state (Bocskocsky, Ezekowitz, & Stein, 2014). These circumstances include longer shot distances, facing tighter defense, increased likelihood of taking their team's next shot, and overall attempting more difficult shots. The same study also determined that players in a "hot" state are more likely to make their next shot after controlling for shot difficulty.

Basketball philosophy has also evolved substantially since Gilovich's analysis in 1985, when offensive play was largely characterized by interior scoring. As player skills have evolved, teams have increasingly emphasized perimeter play and distance shooting (Wang & Zheng, 2022). In addition, factors such as pace of play may further influence shooting patterns. Together, these changes in offensive strategy and shot selection suggest that conclusions drawn from earlier periods may not fully generalize to the context of modern basketball, potentially altering the interpretation of the hot-hand effect.

Ultimately, the presence of the hot-hand phenomenon has largely remained unsolved due to differences in empirical research and statistical examinations of simulated data, which often rely on distinct assumptions and methodologies (Bar-Eli et al., 2006). Additionally, no clear definition of the hot hand exists, with studies varying in how they define and measure streak-shooting behavior. These discrepancies have contributed to ongoing debate regarding the validity of the effect.

2.3 Motivation

Many hot-hand analyses have been conducted in controlled settings or within specific game contexts, such as the NBA Three-Point Contest (Ross, 2017; Lantis & Nesson, 2024) or free throw shooting (Arkes, 2010; Chang, 2019; Lantis & Nesson, 2021). Many of these foundational studies were also conducted in a different era of the NBA (Gilovich et al., 1985), when the style of play differed substantially from today's game. These environments offer valuable insights in their respective contexts, but may differ in applicability to modern in-game playstyles.

The NBA has undergone notable changes in offensive philosophy from past eras, with a greater emphasis on perimeter shooting, increased pace of play, and the development of more versatile offensive skill sets. As a result, modern gameplay reflects a substantially different environment than previous eras. Therefore, reexamining the hot hand using modern, in-game data provides a new perspective on the presence and magnitude of streak shooting. Specifically, this study examines the hot-hand effect across players and seasons in the modern NBA, with the goal of evaluating streak-shooting tendencies over the past 10 seasons.

Chapter 3

DATA

3.1 Modern NBA Trends

All data and figures presented in this section are derived from publicly available NBA statistics obtained from Basketball Reference (Sports Reference LLC. Basketball-Reference.com - Basketball Statistics and History. https://www.basketball-reference.com/. Accessed Oct 15, 2025).

The most notable example of the evolution of offensive strategy is Stephen Curry's 2015-16 unanimous MVP season, in which he attempted a then-record 886 three-pointers. Figure __ below displays an increasing trend in the percentage of total field goal attempts (FGA) that were three-point attempts (3PA) after the 2015-16 regular season. The season variable is encoded as the ending year of each NBA season (e.g., 2016 corresponds to the 2015-16 season).

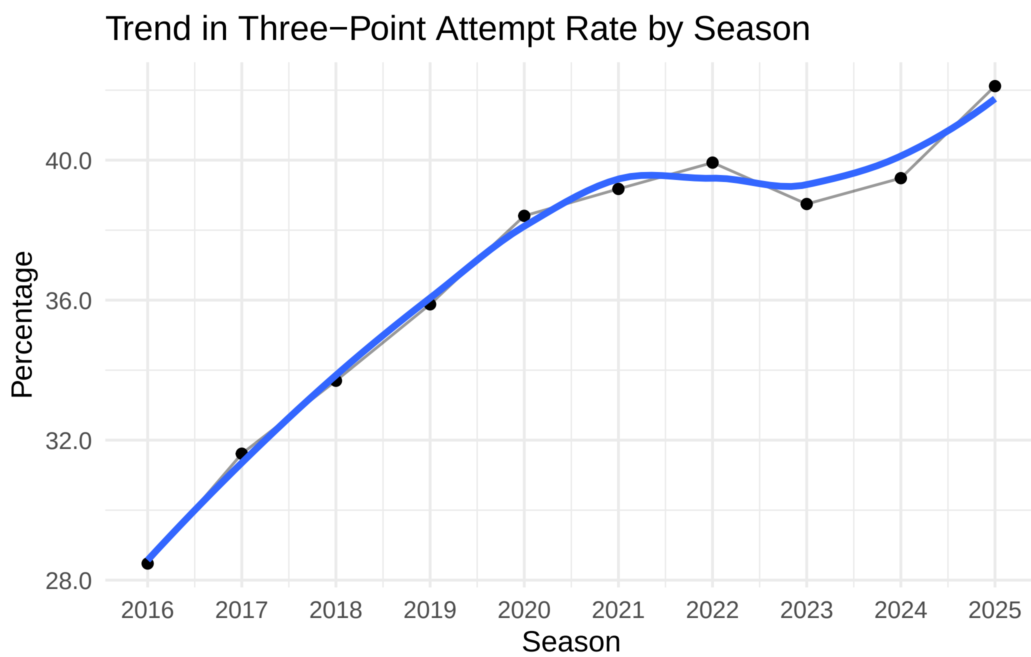

Figure 3.1: Increasing trend in three-point share by season.

The increase in three-point shooting raises the question of whether the hot-hand effect has evolved over time. Improvements in offensive skill could amplify streakiness, whereas the lower success rate of three-point attempts relative to two-point attempts may attenuate such effects.

Since the 2015-16 regular season, teams have also emphasized a faster pace of play to get more offensive possessions in a game. Pace, defined as the number of possessions a team uses in a standard 48-minute game, heavily influences the number of shots taken within a game and could have an impact on streak shooting. Figure __ below depicts a marked increase in the league-average pace from the 2015-16 to 2019-20 regular season, followed by a gradual decline and stabilization in recent seasons.

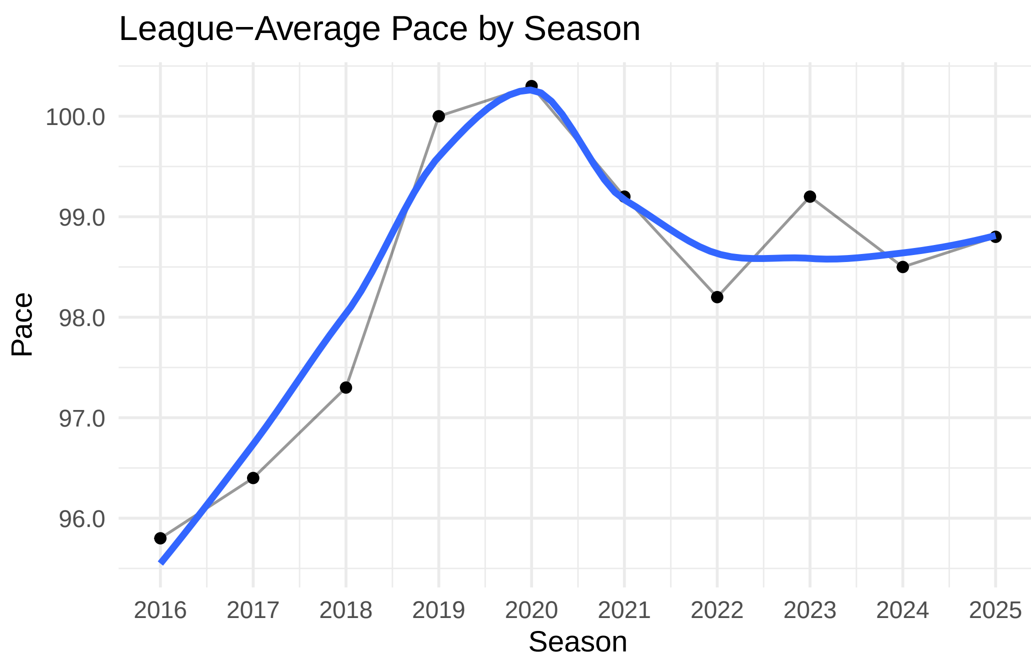

Figure 3.2: League-average pace varies by season, experiencing rapid change after 2016.

The fluctuations in pace since 2016 may suggest changes in streak-shooting tendencies over time, as players attempt more shots per game and therefore have more opportunities to develop scoring streaks.

Figure __ displays the league-average field goal attempts (FGA) per game at the team level increasing at a commensurate rate to the pace of play from approximately the 2015-16 to the 2019-20 regular season. Field goal attempts are defined as "any live-ball throw, shot, or tap—including layups and dunks—directed at a player's own basket to score, excluding free throws" (National Basketball Association, "Rule No. 4: Definitions," *official.nba.com*, accessed Oct 15, 2025).

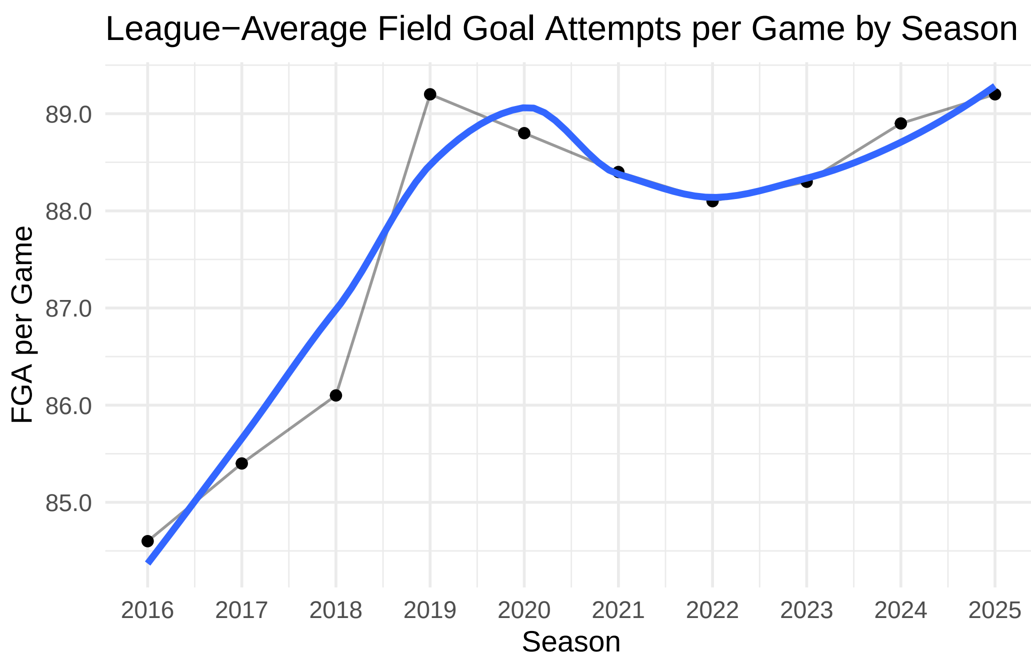

Figure 3.3: League-average FGA per game at the team level increasing after 2016.

The increase in field goal attempts per game coincides with the increase of pace, suggesting a fundamental alteration in offensive philosophy that may influence streak-shooting tendencies.

3.2 Data Collection

3.2.1 Sample Selection

Data on the top 50 players with the most field goal attempts per season were collected for each of 10 seasons from 2015-16 to 2024-25 (not including playoffs), yielding a total of 500 player-seasons. A player-season is defined as a single season of performance for an individual player (e.g. 2015-16 Stephen Curry, 2019-20 LeBron James). Restricting the sample to the top 50 shooters per season focuses the analysis on high-volume players, who account for a large share of their team's shot attempts and are often among the most influential contributors. This approach reduces the overall sample size of the dataset while preserving a high number of shot attempts, allowing for higher statistical power in detecting a hot-hand effect. Importantly, some players were top 50 shooters in FGA in multiple seasons from

2015-16 to 2024-25. These players are represented multiple times in the analysis with unique shot sequences for each respective season in which they were a top 50 shooter. The 2015-16 season was chosen as the starting point due to its significance in changing league-wide trends in shot selection. It is of interest to analyze how the hot hand has progressed since that season, as more teams attempt to replicate the success of prolific offensive teams from that time frame.

All data were collected using the R package hoopR (Gilani, 2023), which provides access to NBA play-by-play statistics through SportsDataverse API. Shot sequence data were obtained for the 500 player-seasons in consideration, with each sequence recording the sequential outcomes of all field goal attempts by a player within a player-season. For example, a shot sequence of ‘0011’, where 0’s represent missed shots and 1’s represent made shots, corresponds to a sequence in which a player missed the first two shots and made the last two shots. A missed shot was also defined as a “failure”, whereas a made shot was defined as a “success”. In total, data from 35,428 games across the 500 player-seasons was collected.

3.2.2 Exploratory Data Analysis

The distribution of the total number of field goal attempts per season for the top 50 shooters is displayed in Figure __ below. Notably, the 2019-20 and 2020-21 seasons exhibit lower overall field goal attempts relative to other seasons, which is likely attributable to COVID-19-related restrictions that reduced the number of games played. Prior to those seasons, the distributions of field goal attempts are bimodal, with the larger peaks occurring at approximately 1,000 field goal attempts and the smaller peaks under 1,500 field goal attempts. After the 2019-20 and 2020-21 seasons, the distributions become more unimodal, with peaks shifting to greater than 1,000 field goal attempts. This pattern could suggest a delayed impact of the 2015-16 season, when long-distance shooting and faster pace became the primary emphasis for most teams. In general, we see an increase in field goal attempts post-COVID, which could alter hot-hand trends.

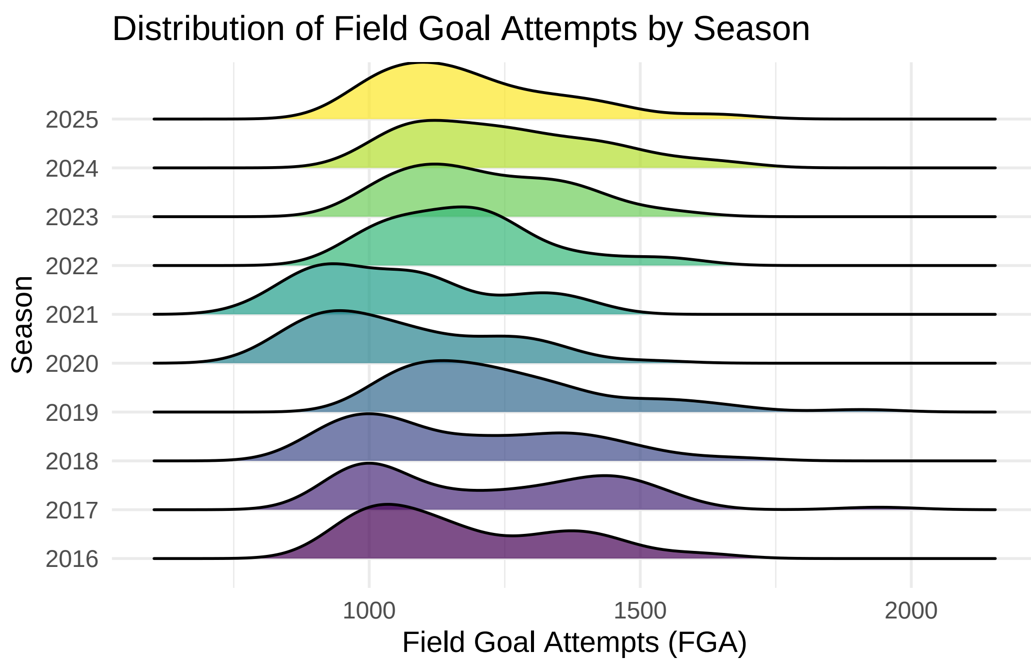

Figure 3.4: Distributions of FGA per game vary by season; shortened seasons in 2020 and 2021.

For each shot sequence, multiple variables were collected or constructed to help analyze the hot-hand effect. We define an *opportunity* as a method of determining whether a player is currently on a hot streak. Specifically, an opportunity is any shot preceded by at least k consecutive made shots. In this analysis, $k = 3$ is used to remain consistent with prior hot-hand literature.

For example, when $k = 3$, consider the sequence:

$$0, 1, 1, 1, \underline{1}, \underline{0}, 1, 1, 1, \underline{1}$$

Shots that occur immediately after three consecutive made shots are classified as opportunity shots and are underlined in the example above. These opportunity shots are used to evaluate player performance while in a "hot" state.

An initial analysis of the data reveals that the mean number of opportunities per game for qualified player-seasons fluctuates across seasons, albeit at small increments, as shown in Table __ below. This trend does not necessarily imply any increase in hot-hand shooting ability, as the increase could stem from a variety of factors. As discussed earlier, league-wide average field goal attempts per game have also increased over time, which could lead to a higher number of observed opportunities simply due to a greater volume of shots being taken.

Table 3.1: Number of Opportunities per Game per Player by Season

Season	Mean Opportunities
2016	1.12
2017	1.34
2018	1.29
2019	1.36
2020	1.31
2021	1.47
2022	1.36
2023	1.53
2024	1.58
2025	1.42

To adjust for the for the increase in the number of field goal attempts taken per game, the percentage of total field goal attempts that were opportunity shots was calculated and plotted in Figure __. The percentage of opportunity shots shows an overall upward trend across the period, but with noticeable year-to-year fluctuations. This increasing trend suggests a higher prevalence of streak occurrences over the past 10 seasons, which could be attributed to changes in offensive strategies or players becoming more capable of entering hot streaks. Importantly, this trend only describes a slight increase in the percentage of opportunity shots as opposed to an actual hot-hand effect.

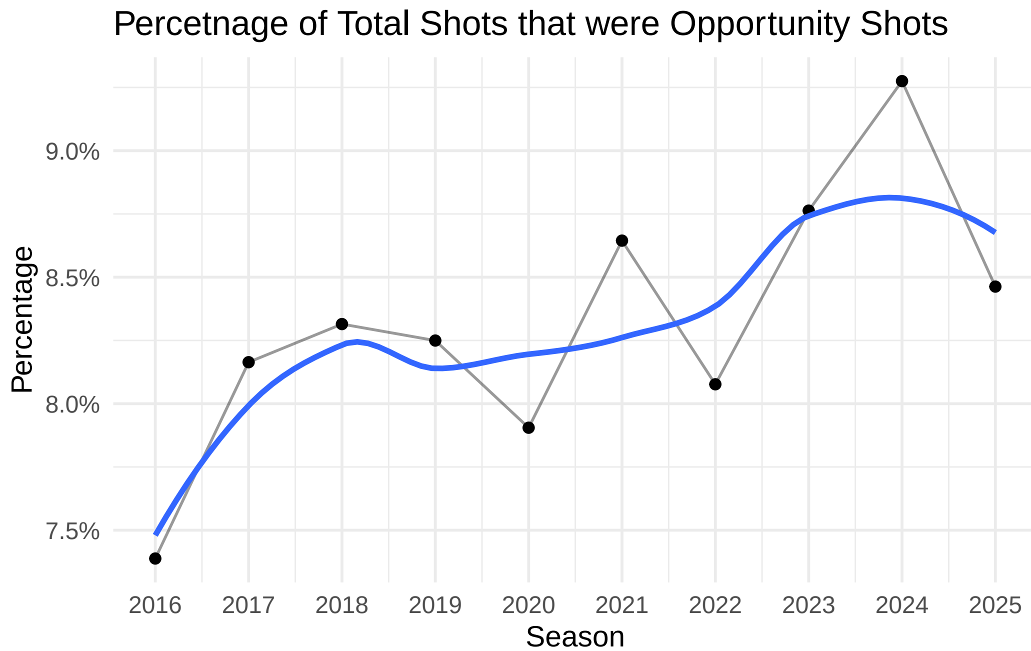

Figure 3.5: Proportion of Opportunity Shots showing general upward trend for qualified player-seasons.

Opportunity field goal percentage, defined as the field goal percentage on opportunity shots, for qualified player-seasons has also generally increased over time, as shown in Figure __. The graph starts at its lowest point during the 2015-16 season and increases steadily until the 2018-19 season, where it initially peaks. The 2019-20 season displays a steep decline in hot hand field goal percentage, but eventually recovers and continues to rise until it peaks during the 2023-24 season. The wave-like pattern of the smoothed line suggests variability over time rather than a strictly monotonic increase. An upward trend provides some preliminary evidence of an increase in hot-hand shooting ability over time, but needs to be formally tested for statistical significance as it may be attributable to random variation.

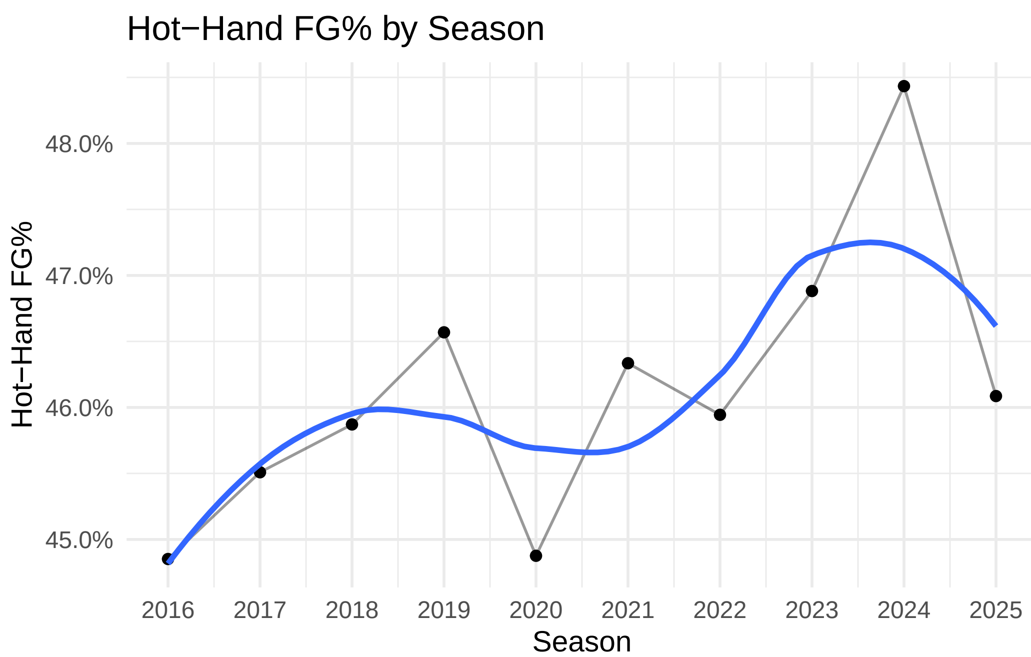

Figure 3.6: Hot-hand FG% showing general upward trend over time for qualified player-seasons.

Although the trends discussed above may indicate a higher degree of streakiness, a meaningful evaluation requires a comparison to baseline shooting performance. Figure __ below addresses this by plotting the difference between opportunity FG% and baseline FG% for all qualified player-seasons from 2015-16 to 2024-25. For each season, the baseline FG% is computed by summing the total number of made field goals from qualified players during that season divided by the total number of field goal attempts during that season from qualified players. This provides a natural benchmark with which the opportunity FG% can be evaluated.

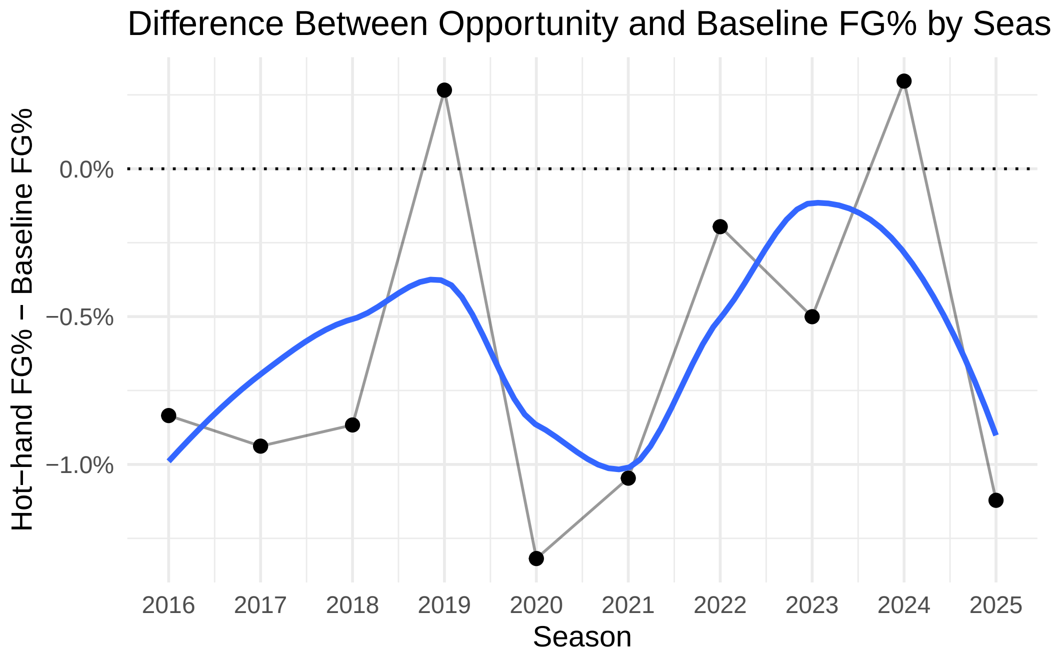

Figure 3.7: Difference between Opportunity and Baseline FG%, with horizontal marker at 0%.

The overall difference seems to fluctuate around zero, with most seasons possessing a slight negative difference, indicating that the success rate on opportunity shots is less than the success rate for baseline attempts on average. While there are isolated seasons, such as 2018-19 and 2023-24, in which hot-hand FG% is greater than the baseline FG%, these deviations are neither persistent nor large in magnitude. These results do not provide any preliminary evidence that a systematic hot hand effect exists on the aggregate level. It is important to note that this analysis remains subject to the selection bias identified by Miller and Sanjurjo, as the hot-hand FG% is computed by conditioning on prior shot outcomes. Therefore, more rigorous and detailed analysis methods that account for this bias are needed before any conclusions can be made regarding the hot-hand effect.

The distance of each shot attempt was collected using hoopR's coordinate system, with each shot containing an x and y-coordinate relative to the basket. Shot distances were then organized into one of three categories: close range (0-8 feet), mid-range (8-22 feet), and three-point (22+ feet). These category labels are approximate, as the distance of a three-point attempt varies depending on court location. For example, corner three-point attempts are taken from approximately 22 feet, whereas three-point attempts taken from the top of the arc must be at least 23.75 feet. The chosen cutoff values reflect the empirical distribution of shot distances, which tend to cluster in specific regions. Figure __ below displays the distribution of shot distances for all games within the 500 qualified player-seasons, with vertical lines marked at cutoff values of 8 and 22 feet.

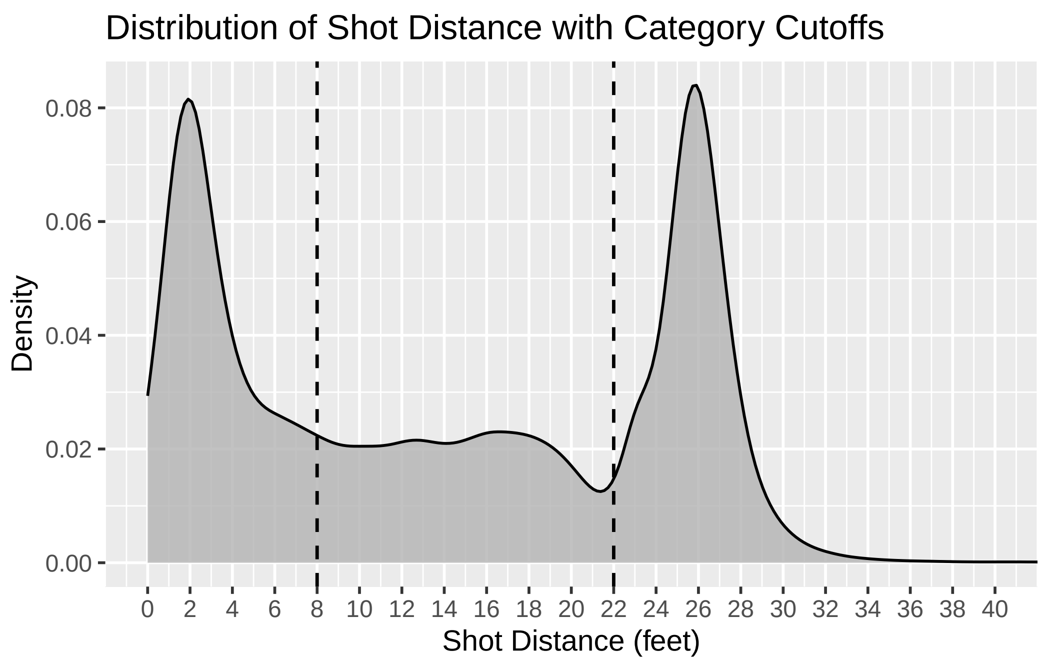

Figure 3.8: Distribution of Shot Distances with cutoff values at 8 and 22 feet.

The density plot exhibits clear clustering of shots near the basket (approximately 0-8 feet) and beyond the three-point line (approximately 22-30 feet), with relatively fewer attempts in the mid-range. This varying structure suggests that teams tend to compartmentalize shot types into various categories rather than a continuous spectrum. Therefore, modeling shot distance as a categorical predictor with three distinct levels provides a more practical and interpretable representation of shot selection than treating it as a purely continuous quantitative variable.

3.3 Streak Statistics

Ross (2017) outlines various *streak statistics* that analyze finite sequences of binary outcomes. A streak statistic is a measure designed to quantify the "streakiness" of a given shot sequence. The streak statistic used in my analysis is defined below:

> *Difference in proportion of S (opportunities - non-opportunities)*: The difference between the proportion of success during opportunity shots and the proportion of success during non-opportunity shots. In the previous example, 0, 1, 1, 1, $\underline{1}$, $\underline{0}$, 1, 1, 1, $\underline{1}$, the value of the streak statistic was $\frac{2}{3} - \frac{6}{7} \approx -0.1905$ (Ross, 2017).

This statistic was chosen because it provides a direct comparison of shooting performance in a "hot" state against shooting performance outside of that state. Additionally, it has an intuitive interpretation under the assumption of independence, as it is often expected to be equal to zero. However, as discussed earlier, this expectation is biased downward in finite samples, which will be revisited in subsequent analysis.

Chapter 4

PERMUTATION TESTING

4.1 Permutation Testing Procedure

The permutation testing procedure utilizes a streak statistic to assess the likelihood of observing a shot sequence at least as extreme as that computed from the observed shot sequence. We define the null and alternative hypotheses:

H_0: Shots are independent.

H_a: Shots are dependent.

The steps of the procedure are outlined below:

1. Calculate the streak statistic for the observed shot sequence.
2. Randomly permute the outcomes of the observed shot sequence while preserving the total number of successes and failures. This produces a sequence consistent with the null hypothesis of independent shot attempts.
3. Calculate the streak statistic for the permuted shot sequence.
4. Repeat steps 2-3 many times (e.g. 10,000 iterations) to approximate the sampling distribution of the streak statistic assuming independence of shot attempts.
5. Calculate the proportion of permuted streak statistics that are at least as extreme as the observed statistic in the direction consistent with the hot-hand effect to obtain the *p*-value.

This procedure addresses the bias identified in Gilovich et al. (1985) by directly constructing an appropriate null distribution from the observed shot sequence rather than relying on theoretical expectations under independence. The random permutation of observed outcomes satisfies the null assumption of independence, allowing us to compare the observed shot sequence to many sequences simulated under the null hypothesis.

4.2 Season-Level and Within-Game Shot Sequences

The permutation testing procedure was applied to season-long shot sequences as well as within-game shot sequences to assess the robustness of results across different levels of aggregation. Each approach provides distinct advantages that address limitations inherent in the other.

4.2.1 Season-long Data

Each observation in this sample consists of an individual player's sequential shot data for a particular season. That is, for each of the 500 player-seasons, a binary sequence of makes and misses is constructed using all shots taken over the course of the season. Shots are carried over across games, meaning they accumulate over the course of a season without resetting after each game. As a result, streaks can extend across game boundaries. For instance, if a player makes the final two shots of one game and the first shot of the next game, the subsequent shot would be classified as an opportunity. Treating each game separately, as examined in the within-game analysis, effectively censors shot sequences at the end of each game by truncating potential streaks. This could result in weaker evidence of a hot-hand effect by reducing the number of observed opportunity situations and inflating p-values. By allowing sequences to carry over across games, this analysis avoids inflating p-values and maximizes the statistical power under the given context.

The permutation test is applied to each of these season-long sequences, allowing for the evaluation of streak behavior over a larger sample of shots per player. Using season-long shot data allows for greater statistical power due to the large number of opportunities within a given sequence. With more qualifying observations, estimates of conditional shooting probabilities become more stable and less sensitive to random variation, increasing the ability to detect meaningful deviations from the null hypothesis. It also offers a unique perspective on hot-hand shooting, with results applying to prolonged stretches of time rather than only individual games. Significant results for season-long data could indicate a potential carry-over effect, with players experiencing increased levels of play across multiple games given that the player finished the previous game on a streak.

Figure __ (full results for all player-seasons provided in Appendix) displays the permutation test results for the 30 player-seasons exhibiting the strongest evidence of a hot-hand effect.

(**TABLE IN APPENDIX**)

At an $\alpha = .05$ individual significance level, only 10 out of 500 player-seasons exhibit a statistically discernible hot-hand effect. However, when conducting 500 independent hypothesis tests, we would expect approximately 0.5 * 500 = 25 false rejections purely by chance, even if no true hot-hand effect exists. Thus, the observed number of significant results is lower than what we would expect under the null hypothesis of independence. Figure ___ below displays the distribution of p-values for the 500 player-seasons. The distribution is skewed left, with many observations possessing p-values greater than 0.05.

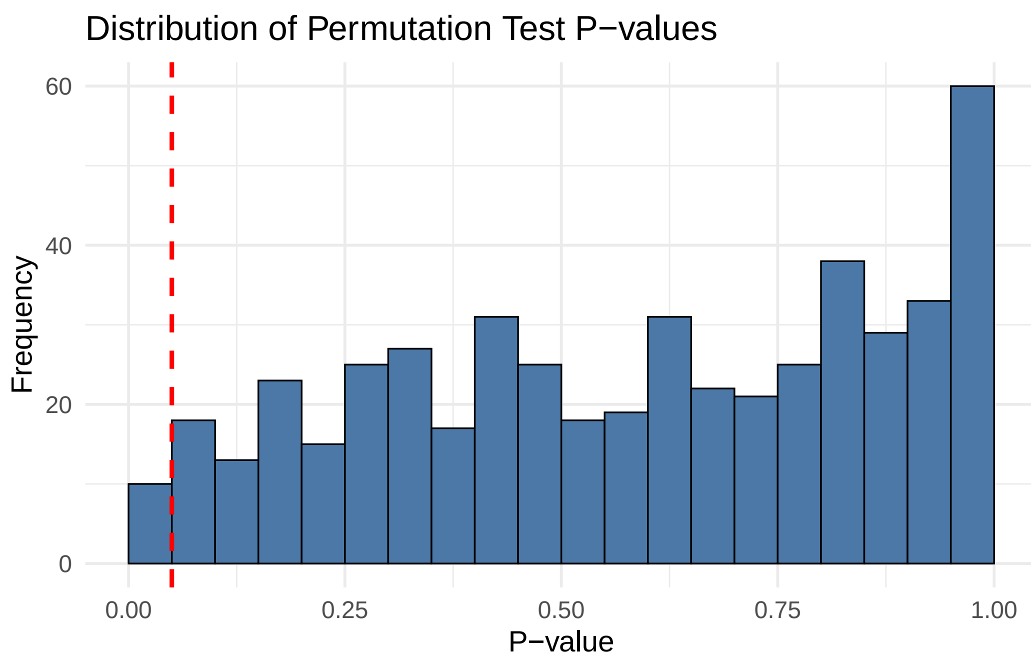

Figure 4.1: Distribution of Permutation Test P-values displaying left skew.

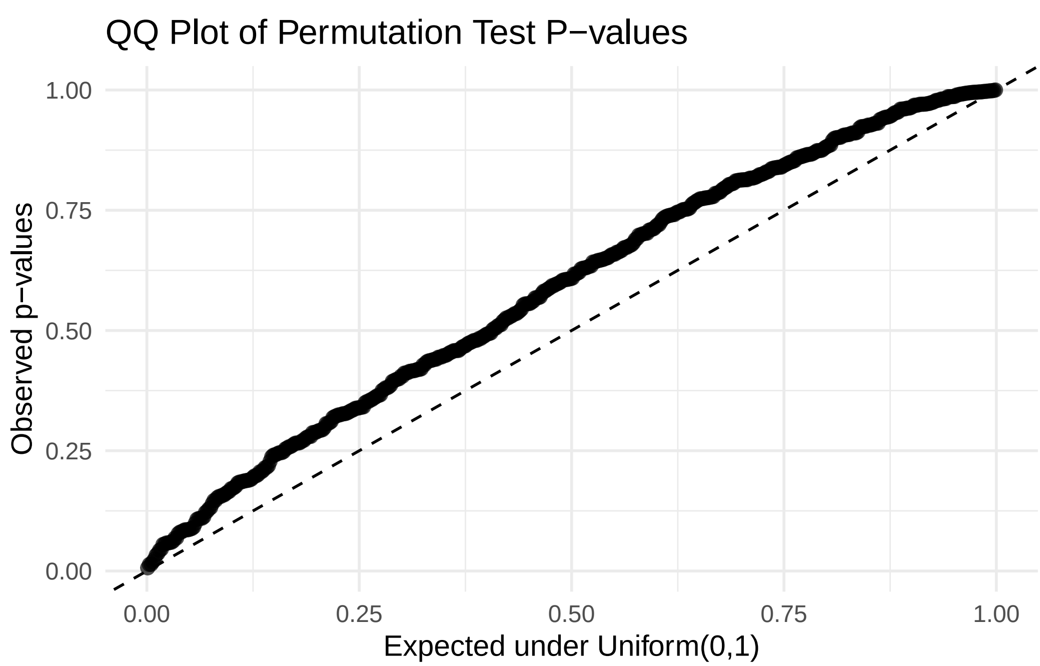

Figure 4.2: Distribution of Permutation Test P-values compared to uniform dsitribution.

The QQ Plot (Figure ___) above compares the observed p-values from the permutation tests to the uniform distribution. Under the null assumption of independence, we expect the distribution of p-values to follow a uniform distribution, displayed by the diagonal line. The hot-hand effect suggests that the distribution of observed p-values should fall below the diagonal line since a larger proportion of player-seasons would be expected to show stronger evidence than under independence, resulting in lower quantiles. However, the observed p-values align above the diagonal line, which suggests that there is not only little evidence of a hot-hand effect, but also that players potentially perform slightly worse during opportunities than other shots. These results are aggregated at the player-season level, based on season-long shot sequences rather than game-level data. As a result, the analysis captures within-season patterns but may mask potential short-term dynamics.

To account for the false discovery rate of hot-hand effects, the Benjamini-Hochberg procedure was applied to the existing p-values. After this adjustment, none of the player-seasons remain statistically significant, providing no evidence of a hot-hand effect after correcting for multiple comparisons.

4.3 Game-level Data

Season-long shot sequences were then deconstructed further into game-level sequences. In this sample, each observation corresponds to a binary sequence of makes and misses for a game within a player-season. Thus, each player-season contributes multiple sequences, one for each game in which the player recorded field goal attempts. The permutation test is applied to each game-level shot sequence separately, capturing within-game streak behavior and allowing for a more granular analysis of hot-hand shooting patterns. Results for game-level data are often considered more practical for hot-hand analysis than season-long data since most people attribute the hot-hand effect to individual games rather than prolonged periods of time. However, using game-level data decreases the statistical power of the permutation test considerably, as shot sequences typically contain very few opportunities, if any.

In total, 35,428 shot sequences were considered for the game-level analysis. The distribution of total FGA per game for qualified players is displayed below in Figure ___, with a mean of 16.50 FGA per game and a standard deviation of 5.19 FGA per game.

Table 4.1: Total Shots per Game per Player

Minimum	Q1	Median	Mean	SD	Q3	Maximum
1	13	16	16.5	5.19	20	50

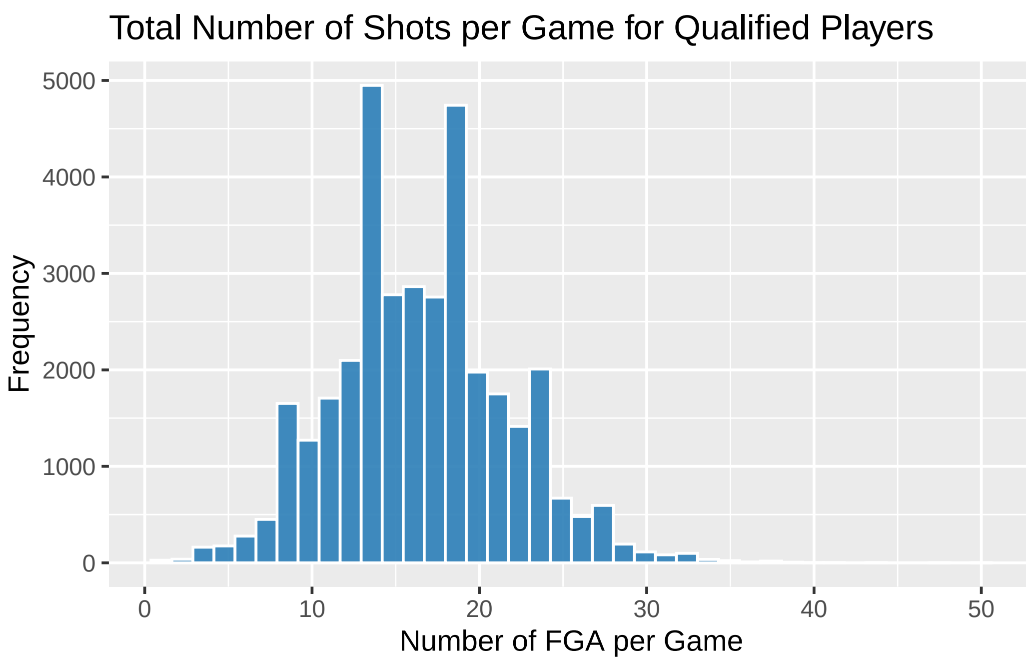

Figure 4.3: Total shots per game for qualified player-seasons.

The limited sample size of each shot vector restrains the number of opportunity shots within a sequence, thereby reducing the statistical power of the permutation test. In addition, 14,563 games (approximately 41.1%) in the sample contain no opportunity shots which further reduces the effective sample size to 20,865 games. Games with zero opportunity shots were retained in the dataset but classified as providing no information for the permutation test.

4.4 Quantifying the Bias

Prior work has shown that the magnitude of small-sample bias decreases as sample size increases (Miller & Sanjurjo, 2018). Thus, the magnitude of bias for the season-long data is minimal and has little effect on the observed streak statistic. In contrast, the limited sample size of shot sequences in the within-game analysis warrant a closer examination of this bias. Because the magnitude of the bias is not immediately clear, a simulation-based approach was used to quantify its impact under realistic conditions for the within-game analysis.

4.5 MCMC Simulation

A Monte Carlo simulation was used to approximate the expected value of the streak statistic *Difference in proportion of S (opportunities – non-opportunities)* under different sequence lengths and success probabilities. A streak length of three was used for the simulation. The estimation procedure is summarized in Algorithm 1.

INSERT ALGORITHM HERE

Figure __ below displays the expected value of the streak statistic using Monte Carlo simulation. The parameter p varies from 0.40 to 0.60, in increments of 0.05, reflecting plausible shooting proportions observed in game settings, although more extreme values exist due to small sample sizes. The overall sample proportion of made shots across all 35,428 qualified games was approximately 0.467. For each (n, p) pair, the simulation was conducted using 10,000 trials and the resulting expected values are presented in the figure below.

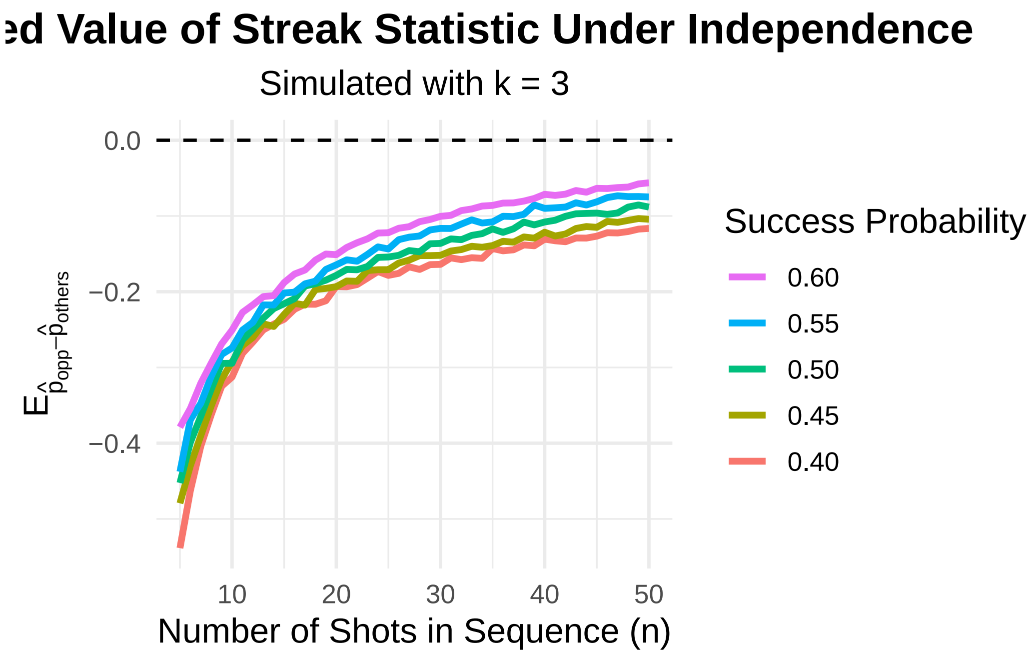

Figure 4.4: Simulated bias of streak statisic under different conditions.

The curves exhibit a consistent vertical ordering with higher success probabilities corresponding to a smaller magnitude of bias. Differences in bias across success probabilities appear relatively consistent, indicating that the bias generally scales proportionally with p. Additionally, each line appears to be monotonically increasing towards 0, meaning that the expected bias decreases as n increases for a given success probability p.

These results can be compared to the null statistics of each observed shot sequence generated by the permutation test. However, it is important to note that shot sequences typically vary with p and n, which would alter the magnitude of bias for that particular sequence.

4.6 Bias-Adjusted Streak Statistic

The permutation test was run for all shot sequences in the sample. Sequences with zero opportunities were assigned a p-value of NA. Table __ in the Appendix contains a sample of 30 games and the collected statistics through the permutation tests. To visualize the effect of different sample sizes on the null distribution, three games of different sample sizes were collected. The three selected games contained 13, 16, and 20 shots to represent Q1, the median, and Q3, respectively. Each game had a streak statistic of -0.5 for consistency. The null distributions along with their corresponding p-value are shown in Table __ below.

Table 4.2: Bias-Adjusted Streak Statistic for Different Sample Sizes

Player	Game ID	Total Shots	Total Makes	Opportunity Percentage	Difference in Proportions	Null Mean	Adjusted Difference in Proportions	P-value
Kyle Kuzma	401468533	20	9	0	-0.5	-0.235	-0.265	0.999
Tobias Harris	400975574	16	7	0	-0.5	-0.242	-0.258	1.000
Jordan Clarkson	401267657	13	6	0	-0.5	-0.292	-0.208	0.969

The *Difference in Proportions* column represents the streak statistic, defined as the difference between the proportion of makes on opportunity shots and the proportion of makes on all other shots. However, this statistic is subject to the bias discovered by Miller & Sanjurjo (2018). The *Adjusted Difference in Proportions* column corrects this statistic by subtracting the null mean obtained from the observed streak statistic. This adjustment accounts for bias in the original streak statistic, yielding a bias-adjusted streak statistic.

As expected, the null means generally seem to decrease as the number of shots increase, although some trend deviation remains due to differences in field goal percentages across games (Figure __, Appendix). The average null mean across all games with at least one opportunity was approximately -0.229, indicating a substantial bias in the streak statistic.

This estimated bias agrees with the results obtained from the Monte Carlo simulation, as our overall success probability was approximately 0.467 and the average number of shots taken per game was approximately 16.5. The null standard deviations are relatively large compared to the season-long analysis, reflecting the high variability inherent in short shot sequences. Because many games contain a limited number of opportunities, the field goal percentage on those opportunities can vary substantially, leading to highly volatile streak statistics. For example, in games with only one opportunity, the opportunity field goal percentage can take on only two values (0 or 1). As a result, the streak statistic is highly discrete, and the corresponding permutation distribution is characterized by a small set of possible values that can differ markedly from one another, contributing to considerable variability in the null distribution. Consequently, games with few opportunities will also have lower statistical power, making it more difficult to detect meaningful hot-hand trends.

4.7 Game-Level Results

Overall, only 562 out of 20,865 games (approximately 2.69%) that included at least one opportunity produced a statistically significant streak statistic at the $\alpha = 0.05$ significance level. This proportion is lower than what would be expected under the assumption of independence, providing little evidence in support of a within-game hot-hand effect. Furthermore, the mean of the *Adjusted Difference in Proportions* column across all games with at least one opportunity was approximately -0.0136, indicating that, on average, players performed slightly worse on opportunity shots than would be expected under independence.

Figure __ provides the distributions of the bias-adjusted streak statistic by season. The distributions appear largely consistent across seasons, exhibiting similar bimodal shapes and comparable peak locations. In particular, each season shows two prominent peaks centered at approximately -0.25 and 0.25. The black points denote the seasonal means of the null-adjusted streak statistic, which are generally centered slightly below zero. This aligns with the previous finding that players tend to perform marginally worse on opportunity shots relative to other shots after adjusting for the bias. One exception is the 2018-19 season in which the mean appears slightly above zero, suggesting a small, though likely negligible, positive hot-hand effect. The similarity between the distributions suggests little evidence of systematic seasonal differences in hot-hand effects.

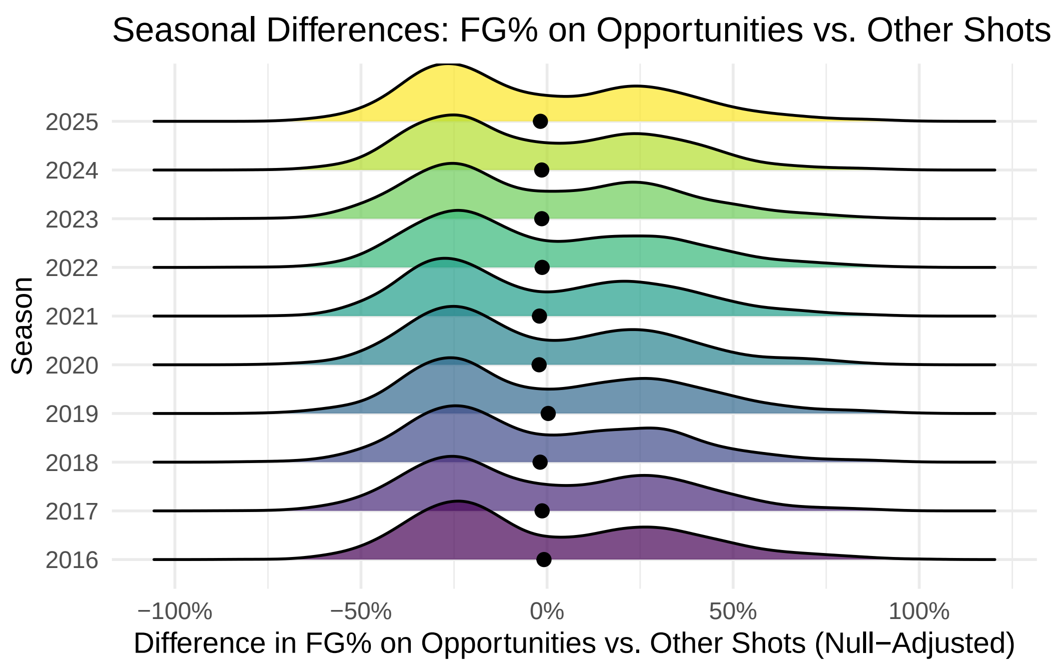

Figure 4.5: Bias-Adjusted Streak Statistic Distributions by Season.

The observed bimodality can be explained by the structure of the streak statistic that results from small shot sequences. Because many games contain few opportunities, the opportunity field goal percentage is often either very low (e.g., 0) or moderate (e.g., around 0.5), while the field goal percentage on other shots is typically near 0.5, reflecting both the larger number of "other" shot attempts in each game and the fact that league-wide field goal percentages tend to be close to this value. Consequently, the unadjusted streak statistic frequently takes values near -0.5 or 0. Figure __ displays this pattern by using the three previously sampled games which contained different numbers of total shots and successful shots. The null distributions reveal points clustering at approximately -0.5 and 0, confirming our previous intuition.

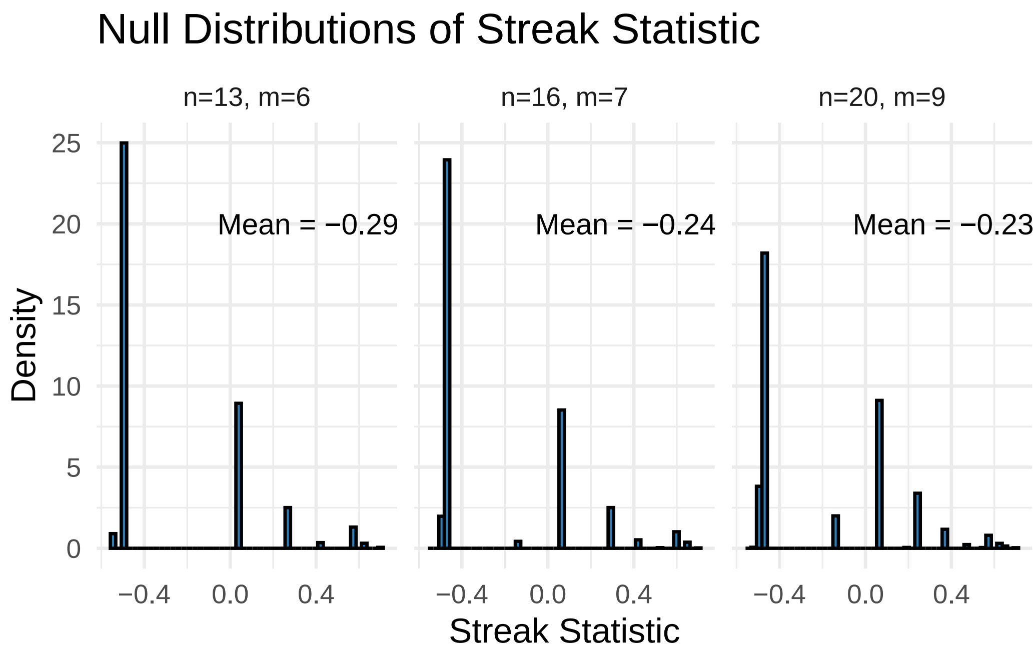

Figure 4.6: Null Distributions of Streak Statistic under Different Scenarios.

After adjusting for the null mean (approximately -0.23), these values shift to around -0.25 and 0.25, producing the observed peaks. This pattern reflects the high variability of the statistic rather than systematic differences across seasons, providing little evidence of a meaningful season-specific hot-hand effect.

Chapter 5

CONCLUSION

Write something here I guess.

5.1 Computational Details

The results in this paper were obtained using `R` 4.4.1. `R` itself and all packages used are available from the Comprehensive `R` Archive Network (CRAN) at https://CRAN.R-project.org/.

REFERENCES

R Core Team. (2024). *R: A Language and Environment for Statistical Computing*. Vienna, Austria: R Foundation for Statistical Computing. Retrieved from https://www.R-project.org/

APPENDICES

Appendix A

SUPER COOL FANCY FUNCTION

```
# Add your fancy function here!
```

Appendix B

SLIGHTLY LESS COOL FANCY SOMETHING

```
# Got more fancy functions, or figures, or tables,
# or pretty much anything?
```